Predicting PVT Behavior, Residual Properties, and Pure-Component Fugacity of Real Gases Using Cubic Equations of State

Indian Institute of Technology Delhi

Bansiwal Ankit
2025CH71777 – Chemical Engg.
ch7251777@chemical.iitd.ac.in

Gupta Rohit Anil
2025CH71467 – Chemical Engg.
ch7251467@chemical.iitd.ac.in

1. Introduction & Generic Cubic EOS Formulation (Milestone 1)

In chemical process engineering, the ideal-gas law ($PV^{ig} = RT$) fails at moderate-to-high pressures because it neglects molecular volume and intermolecular forces (**Smith et al., 2018**). This project develops a Python software package (`pvt_thermo_project.py`) to predict real-gas volumetric behavior (Z, V), residual enthalpy (H^R), residual entropy (S^R), and pure-component gas fugacity (φ, f) across four cubic Equations of State (EOS): van der Waals (**1873**), Redlich–Kwong (**1949**), Soave–Redlich–Kwong (**1972**), and Peng–Robinson (**1976**). Following Section 3.6 of **Smith et al. (2018)**, all four models are unified into a single generic cubic EOS:

$$P = RT / (V - b) - a(T) / [(V + \varepsilon b)(V + \sigma b)] \quad \textbf{(Eq. 1 / SVNA 3.41)}$$

Here, the repulsive term $RT/(V - b)$ accounts for the excluded co-volume b occupied by the molecules themselves, while the attractive term $a(T)/[(V + \varepsilon b)(V + \sigma b)]$ accounts for long-range intermolecular attraction forces that reduce the pressure below its ideal-gas value. Imposing the critical-point inflection constraints $(\partial P/\partial V)_{T,\mathrm{cr}} = 0$ and $(\partial^2 P/\partial V^2)_{T,\mathrm{cr}} = 0$ relates b and $a(T)$ to the critical temperature T_c, critical pressure P_c, reduced temperature $T_r \equiv T/T_c$, reduced pressure $P_r \equiv P/P_c$, and Pitzer acentric factor ω (**Pitzer, 1995**). Introducing dimensionless parameters β and q (Eqs. 3.44–3.51) gives:

$$b = \Omega RT_c / P_c, \quad a(T) = \Psi \alpha(T_r; \omega) R^2 T_c{}^2 / P_c \;\Rightarrow\; \beta \equiv bP / RT = \Omega P_r / T_r, \quad q \equiv a(T) / (bRT) = \Psi \alpha(T_r; \omega) / (\Omega T_r) \quad \textbf{(Eq. 2 / SVNA 3.44–3.51)}$$

Here, β represents the dimensionless molecular co-volume and q represents the dimensionless ratio of attractive to thermal energy. Substituting molar volume $V = ZRT/P$ into Eq. (1) yields the dimensionless equation $Z = 1 + \beta - q\beta(Z - \beta)/[(Z + \varepsilon\beta)(Z + \sigma\beta)]$ (Eq. 3.48), which expands into the canonical cubic polynomial in compressibility factor Z:

$$Z^3 + [(\varepsilon + \sigma - 1)\beta - 1]Z^2 + [\varepsilon\sigma\beta^2 - (\varepsilon + \sigma)\beta(1 + \beta) + q\beta]Z - [\varepsilon\sigma\beta^2(1 + \beta) + q\beta^2] = 0 \quad \textbf{(Eq. 3)}$$

Table 1: Universal Constants and Temperature Functions for Cubic Equations of State (Table 3.1, Smith et al., 2018)

EOS Model	$\alpha(T_r;\,\omega)$	σ	ε	Ω	Ψ	Z_c	$d\ln\alpha / d\ln T_r$
vdW (1873)	1	0	0	0.12500	0.42188	0.37500	0
RK (1949)	$T_r^{-1/2}$	1	0	0.08664	0.42748	0.33333	−0.5
SRK (1972)	$[1 + m(1 - T_r^{1/2})]^2$	1	0	0.08664	0.42748	0.33333	$-m(T_r/\alpha)^{1/2}$
PR (1976)	$[1 + \kappa(1 - T_r^{1/2})]^2$	$1 + \sqrt{2}$	$1 - \sqrt{2}$	0.07780	0.45724	0.30740	$-\kappa(T_r/\alpha)^{1/2}$

2. Derivations of Residual Properties (H^R, S^R) & Gas Fugacity (φ, f) (Milestones 2 & 3)

In Table 1, $m = 0.480 + 1.574\omega - 0.176\omega^2$ (SRK) and $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$ (PR). A residual property $M^R \equiv M - M^{ig}$ quantifies the departure of a real fluid property from its ideal-gas value at the same T and P (Eq. 6.41). Because cubic EOS models are pressure-explicit, the pressure integrals for G^R and H^R are transformed in Section 6.3 (p. 227) of **Smith et al. (2018)** into molar density ($\rho \equiv 1/V$) integrals via $dP/P = d\rho/\rho + dZ/Z$ at constant T:

$$G^R / RT = \int_0^\rho (Z-1)(d\rho/\rho) + Z - 1 - \ln Z, \quad H^R / RT = -T\int_0^\rho (\partial Z/\partial T)_\rho (d\rho/\rho) + Z - 1 \qquad \textbf{(Eq. 4 / SVNA 6.58–6.59)}$$

Following Section 13.6 (pp. 487–489), writing Eq. (1) in terms of ρ yields $Z - 1 = \rho b/(1-\rho b) - q\rho b/[(1+\varepsilon\rho b)(1+\sigma\rho b)]$ and $(\partial Z/\partial T)_\rho = -(dq/dT)\rho b/[(1+\varepsilon\rho b)(1+\sigma\rho b)]$. Substituting $\rho b = \beta/Z$ and integrating from $\rho = 0$ to ρ defines the universal departure integral I:

$$I \equiv \int_0^\rho d(\rho b)/[(1+\varepsilon\rho b)(1+\sigma\rho b)] = [1/(\sigma-\varepsilon)]\ln[(Z+\sigma\beta)/(Z+\varepsilon\beta)] \ \text{(for } \sigma \neq \varepsilon\text{)}; \quad I = \beta/Z \ \text{(for vdW)} \qquad \textbf{(Eq. 5 / SVNA 13.72)}$$

Differentiating $q = \Psi\alpha(T_r;\,\omega)/(\Omega T_r)$ with respect to T at constant pressure gives $T(dq/dT) = T_r(dq/dT_r) = [d\ln\alpha(T_r)/d\ln T_r - 1]q$. Substituting this derivative and integral I into Eq. (4) and applying the fundamental residual Gibbs relation $S^R/R = H^R/RT - G^R/RT$ (Eq. 6.47) yields the exact analytical equations for residual enthalpy and residual entropy:

$$H^R / RT = Z - 1 + [d\ln\alpha(T_r)/d\ln T_r - 1]\,qI, \quad S^R/R = \ln(Z-\beta) + [d\ln\alpha(T_r)/d\ln T_r]\,qI \qquad \textbf{(Eq. 6 / SVNA 13.75–13.76)}$$

In Eq. (6), the term $(Z - 1)$ in H^R/RT accounts for the PV work difference between the real gas and an ideal gas, while the term proportional to qI captures the change in intermolecular potential energy upon isothermal compression. In S^R/R, $\ln(Z - \beta)$ reflects configurational space reduction from excluded volume b, while the qI term accounts for molecular ordering due to attractive forces. Finally, by Eq. (10.33), the pure-species fugacity coefficient $\varphi \equiv f/P$ satisfies $\ln\varphi = G^R/RT = H^R/RT - S^R/R$, yielding the exact pure-component gas fugacity equations:

$$\ln \varphi = G^R / RT = Z - 1 - \ln(Z - \beta) - qI \quad \Rightarrow \quad \varphi = \exp(\ln \varphi), \quad f = \varphi P \qquad \text{(Eq. 7 / SVNA 10.34, 13.85)}$$

3. Software Architecture & Textbook Benchmark Verification

The Python program (`pvt_thermo_project.py`) executes four sequential steps: (1) retrieves pure-species critical constants (T_c, P_c, ω) from the built-in Table B.1 database or user input; (2) computes T_r, P_r, α, $d \ln \alpha / d \ln T_r$, β, and q using R = 83.1446 bar·cm^3·mol^{-1}·K^{-1} and 8.31446 J·mol^{-1}·K^{-1}; (3) solves Eq. (3) via companion-matrix eigenvalues (`numpy.roots`), filtering for real roots with $Z > \beta$ and selecting the largest real root Z_{gas} = max(Z) for the vapor phase to compute $V = ZRT/P$; and (4) evaluates I, H^R, S^R, φ, and f via Eqs. (5–7). Tables 2 and 3 verify the code against two benchmarks in **Smith et al. (2018)** for n-butane (T_c = 425.1 K, P_c = 37.96 bar, ω = 0.200): Example 3.9 (p. 101, saturated vapor at 350 K, 9.4573 bar, experimental V = 2482 cm^3·mol^{-1}) and Example 13.5 / Table 13.7 (p. 490, supercritical gas at 500 K, 50 bar). The software reproduces every textbook value exactly.

Table 2: Benchmark 1 — Saturated *n*-Butane Vapor at *T* = 350 K, *P* = 9.4573 bar (SVNA Example 3.9, p. 101)

EOS Model	Z (gas)	V_{code} (cm^3·mol^{-1})	V_{SVNA} (cm^3·mol^{-1})	H^R (J·mol^{-1})	S^R (J·mol^{-1}·K^{-1})	φ	f (bar)
vdW	0.8669	2667.45	2667	−907.80	−1.559	0.8829	8.350
RK	0.8305	2555.47	2555	−1389.33	−2.663	0.8546	8.082
SRK	0.8191	2520.44	2520	−1580.83	−3.133	0.8467	8.007
PR	0.8081	2486.44	2486	−1599.27	−3.098	0.8378	7.923

Table 3: Benchmark 2 — Supercritical *n*-Butane at *T* = 500 K, *P* = 50 bar (SVNA Example 13.5 / Table 13.7, p. 490)

Method / EOS	Z (gas)	V (cm^3·mol^{-1})	H^R_{code} (J·mol^{-1})	H^R_{SVNA} (J·mol^{-1})	S^R_{code} (J·mol^{-1}·K^{-1})	S^R_{SVNA} (J·mol^{-1}·K^{-1})	φ	f (bar)
vdW	0.6608	549.42	−3937	−3937	−5.424	−5.424	0.7445	37.224
RK	0.6850	569.53	−4505	−4505	−6.546	−6.546	0.7434	37.171
SRK	0.7222	600.47	−4824	−4824	−7.413	−7.413	0.7642	38.208
PR	0.6907	574.28	−4988	−4988	−7.426	−7.426	0.7359	36.794

4. Parametric Case Study: Methane at 350.0 K (1–100 bar) & Conclusions

Table 4 and Figure 1 present the parametric evaluation of Methane (CH_4: T_c = 190.6 K, P_c = 45.99 bar, ω = 0.012) at T = 350.0 K (T_r = 1.836) from 1 to 100 bar using the Peng–Robinson EOS. At 1 bar, methane behaves almost ideally (Z = 0.9991, φ = 0.9991, f = 0.999 bar). As pressure rises to 100 bar, intermolecular attractions draw molecules closer together, reducing Z to 0.9102 and lowering the effective thermodynamic escaping pressure to f = 89.84 bar (φ = 0.8984) compared to the ideal line $f = P$. Both H^R (−1.283 kJ·mol^{-1} at 100 bar) and S^R (−2.771 J·mol^{-1}·K^{-1} at 100 bar) decrease monotonically below zero because attractive molecular interactions lower internal potential energy and restrict spatial disorder relative to an ideal gas.

Table 4: Parametric Evaluation of Methane (CH_4) at T = 350.0 K from 1 to 100 bar Using the Peng–Robinson EOS

P (bar)	P_r	Z (gas)	V ($cm^3 \cdot mol^{-1}$)	H^R ($kJ \cdot mol^{-1}$)	S^R ($J \cdot mol^{-1} \cdot K^{-1}$)	φ	f (bar)
1.0	0.0217	0.9991	29073.18	−0.013	−0.029	0.9991	0.999
20.0	0.4349	0.9806	1426.79	−0.264	−0.584	0.9803	19.606
40.0	0.8698	0.9613	699.34	−0.527	−1.163	0.9602	38.409
60.0	1.3046	0.9426	457.18	−0.787	−1.727	0.9398	56.390
80.0	1.7395	0.9254	336.61	−1.040	−2.266	0.9191	73.528
100.0	2.1744	0.9102	264.88	−1.283	−2.771	0.8984	89.842

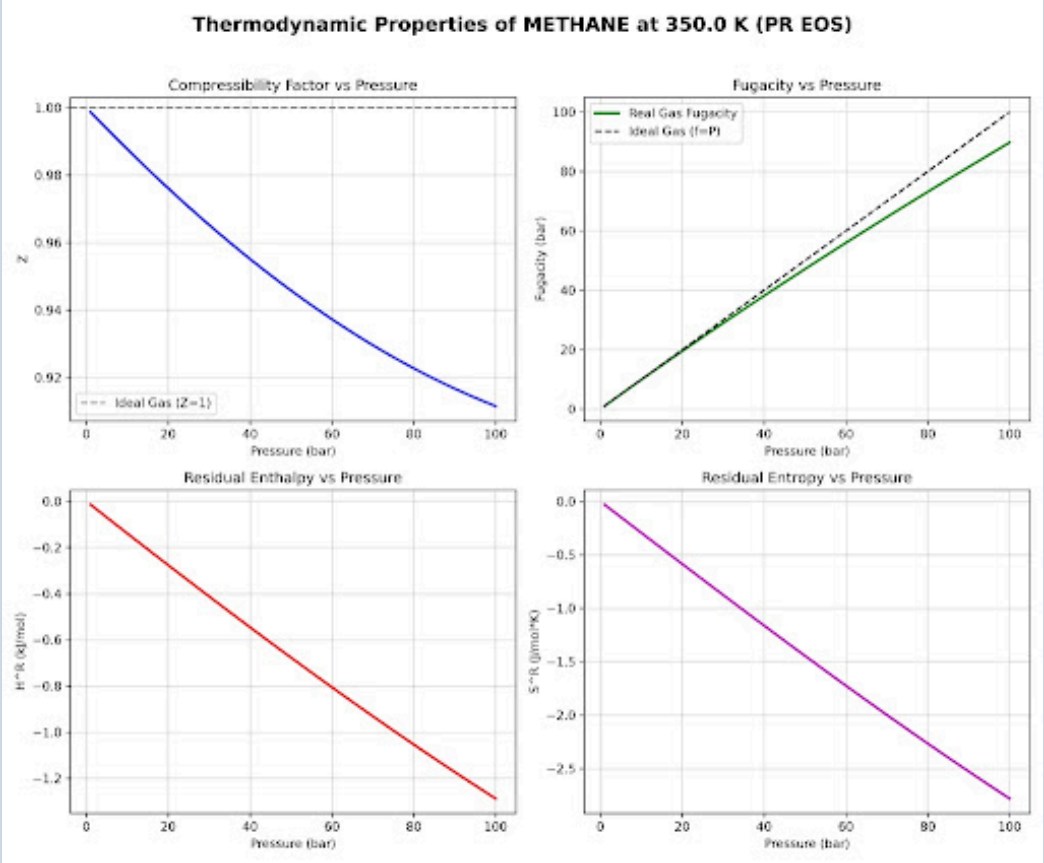


Figure 1: Variation of compressibility factor Z, pure-component gas fugacity f, residual enthalpy H^R, and residual entropy S^R with pressure for Methane at 350.0 K computed using the Peng–Robinson EOS.

5. References

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